A Quantitative Analysis of Air Quality in Harrisonburg Virginia

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1 Introduction

In this paper, we present both a qualitative and quantitative set of arguments to show that in the city of Harrisonburg, air pollutants such as PM 2.5 and AQI have higher density in regions near roads and city life, PM 2.5 is strongly correlated with AQI, and although temperature in Harrisonburg varies widely in the spring as the weather changes, there does not exist a significant correlation with air pollutants such as PM 2.5.

1.1 Background

The Air Quality Index, or AQI, is a numerical measurement of how clean the air is at a given location. The lower the AQI, the cleaner the air. Airborne contaminants can have serious effects on one's health, particularly with repeated or long-term exposure. In response to these risks, the EPA has placed limits on dangerous pollutants. For example, PM 2.5 has a legal limit of 9 micrograms/cubic meter averaged over a year (U.S. EPA, 2024a). These limits have been placed through the National Ambient Air Quality Standards. As students in Harrisonburg, the quality of the air should be our concern. If we learn more about these pollutants we can better understand why what we're measuring matters.

Particulate matter is one of the more harmful pollutants. Its small size allows particles to penetrate the lung tissue and, in some cases, enter the bloodstream. Main sources of PM include car exhaust, industrial emissions, and open fires. Construction dust and some atmospheric reactions are also sources (California Air Resources Board, n.d.). The EPA has found particulate matter to irritate the lungs short term, cause heart disease, reduce lung function, and even cause early death in the long term (U.S. EPA, 2024b). PM2.5 has been found to physically damage lung tissue (Xing et al, 2016).

One major cause of poor air quality is traffic. Take, for example, ground-level ozone. Rather than being emitted from the tailpipe, it's formed when emissions react with sunlight. Breathing ground-level ozone can inflame the airways, reduce lung function, and worsen asthma symptoms (U.S. EPA, 2026). The World Health Organization (2013) found that PM2.5 levels within a few hundred meters of major roads were nearly 25 percent higher than background levels. We kept this in mind when choosing our sample sites.

The purpose of this study was to measure PM2.5, PM10, ozone, AQI, and temperature at locations across Harrisonburg and determine whether areas closer to major roads show higher pollutant levels than quieter parts of the city, as well as to determine other correlative relationships between independent variables, as described above. Data was collected by a number of student groups, totaling over 200 sampling points throughout Harrisonburg. With this coverage, we can draw conclusions about air quality citywide.

2 Methodology

Throughout this study, we measured air quality with the PocketLab Air Sensor by traveling around the City of Harrisonburg and recording data for five minutes at a time. Our group collected samples based on both the hypothesis that air quality would be worse near major roads, as well as convenience for transportation to sampling locations. After collecting our data, we conglomerated our samples with those of our classmates, and used both QGIS and SPSS to examine them by making maps, building regression models, and analyzing elementary statistics about our dataset.

Expanding on our data analysis process, we used QGIS to separate our group's collected data points from the rest of the data collected by our classmates, and then made an interpolated surface based on the "non-group" data. We then compared our true recorded values at their respective geographic locations with the predicted values, and built a table to store the values, as well as the absolute error values. Next, we built a second table containing general statistics about our data: median values, maximum values, and minimum values. After finishing our quantitative data analysis in QGIS, we then turned to using SPSS to built correlation plots between PM 2.5 and AQI, TempC, and PM 10, respectively. We interpreted these findings under the assumption of "good data," and used best econometric practices to draw inferences and conclusions about linear relationships between our random variables.

Beyond quantitative analysis, we also constructed heatmaps of PM 2.5 and AQI interpolation to show both the geographical data about air pollution, as well as the correlation of PM 2.5 with AQI. This provided us with a very clear intuition from which to build a qualitative hypothesis on which to rest our quantitative results.

In summary, our group collected data with the qualitative hypothesis in mind that areas near roads would have higher PM 2.5 and AQI values. We then studied the data that we collected qualitatively and quantitatively in order to build an argument for the conclusions stated in our thesis statement, or equivalently, section 4.6.

3 Results

In this section, we will introduce the results of our data collection and provide tables, maps, and graphs which will support our conclusions as stated in both our thesis statement and section 4.6.

3.1 Tables

After separating our data into A/B groups determined by which data our group collected versus the rest of the class, we interpolated a surface of predicted values based on the "non-group PM 2.5 data." In the following table, we compare the true values of PM 2.5 with the predicted values for PM 2.5 at the locations where we collected data. We then state the absolute error of our prediction. Our main finding was that the mean absolute error value for PM 2.5 was 13.8 micrograms per cubic meter.

True PM 2.5	Predicted PM 2.5	Absolute Error
5	26.01	21.01
37	19.71	17.29
35	15.55	19.45
32	14.27	17.73
6	14.20	8.20
33	14.19	18.81
7	12.77	5.77
7	12.31	5.31
33	11.69	21.31
5	8.14	3.14

Table 1

There are no particular absolute error values worth noting besides the mean absolute error; this is a fairly large error term, given strong variance in the true PM 2.5 data. That said, the variance in the true PM 2.5 data seems odd, although, without including geographic context in the table, we are unable to draw strong conclusions about why this phenomenon occurs.

In addition to this analysis of absolute error of interpolated prediction, we extracted basic statistical parameters regarding our data, as shown below in table 2. It is worth noting that 32.2 degrees Celsius in February in Harrisonburg, VA is an extremely uncommon temperature, and is likely a measurement error of some kind. Likewise, the ozone measurement of 36.2 is an extreme outlier coming from group five of section three, and suggests that certain data points were randomly generated and submitted for purpose of obtaining credit, not performing scientific analysis.

Statistic	PM 10	PM 2.5	Ozone	AQI	Temperature (Celsius)
Median	13	11	0.2	42.2	19.8
Maximum	45	37	36.2	104.7	32.2
Minimum	0	0	0	0	-1.5

Table 2

3.2 Maps

Map one shows the collection of all data collection sites, differentiating between our group's collection sites (purple) and the rest of the class's collection sites (green). It is worth noting that our group's collection sites fall into two clear clusters: northern Harrisonburg and southern Harrisonburg. This is essentially due to matters of convenience for collecting data near our apartments. Additionally, it is interesting that northeastern and northwestern Harrisonburg were largely unexplored, and also that any data points collected in the southeast region fell *outside* of the Harrisonburg city limits.

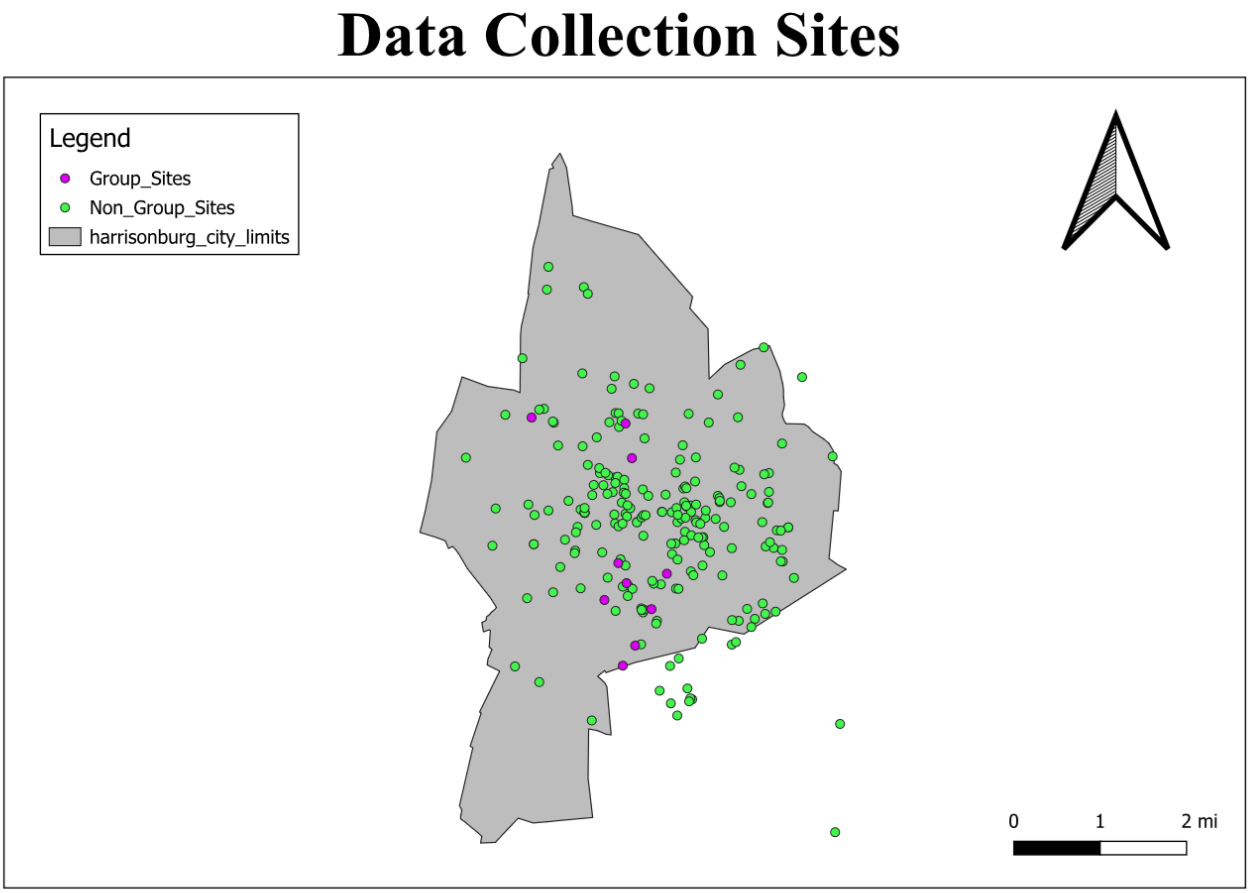

Figure 1: Map 1

Beyond just describing where our data was collected from, we can also compare and contrast the heatmap determined by the interpolation program on PM 2.5 and AQI data. We see very clearly in the maps (two and three respectively) on the next page that higher AQI correlates with higher PM 2.5 geographically, and that many of the "paths of red" in the maps look like roads leading out of the downtown area of Harrisonburg. Additionally, in the western portion of the city limits (in both maps), we see clumps of dark green - corresponding to lower PM 2.5 and lower AQI - which appear near land which has been zoned for farming. This supports the conclusion that farms contribute to better air quality in our region.

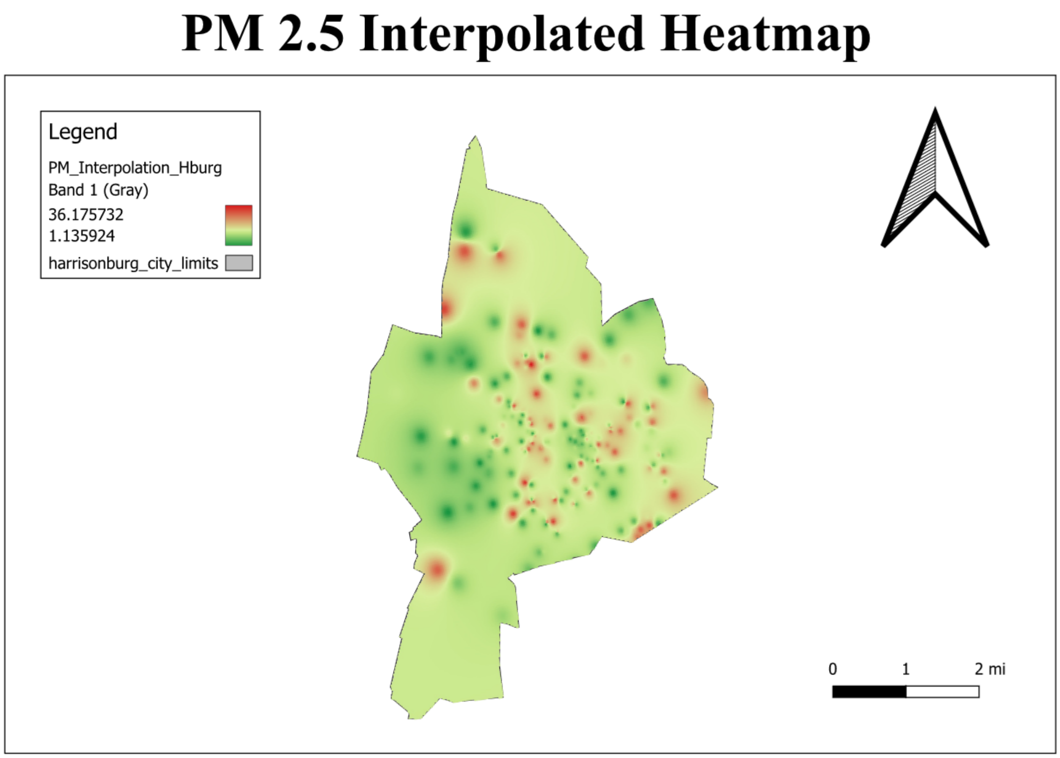

Figure 2: Map 2

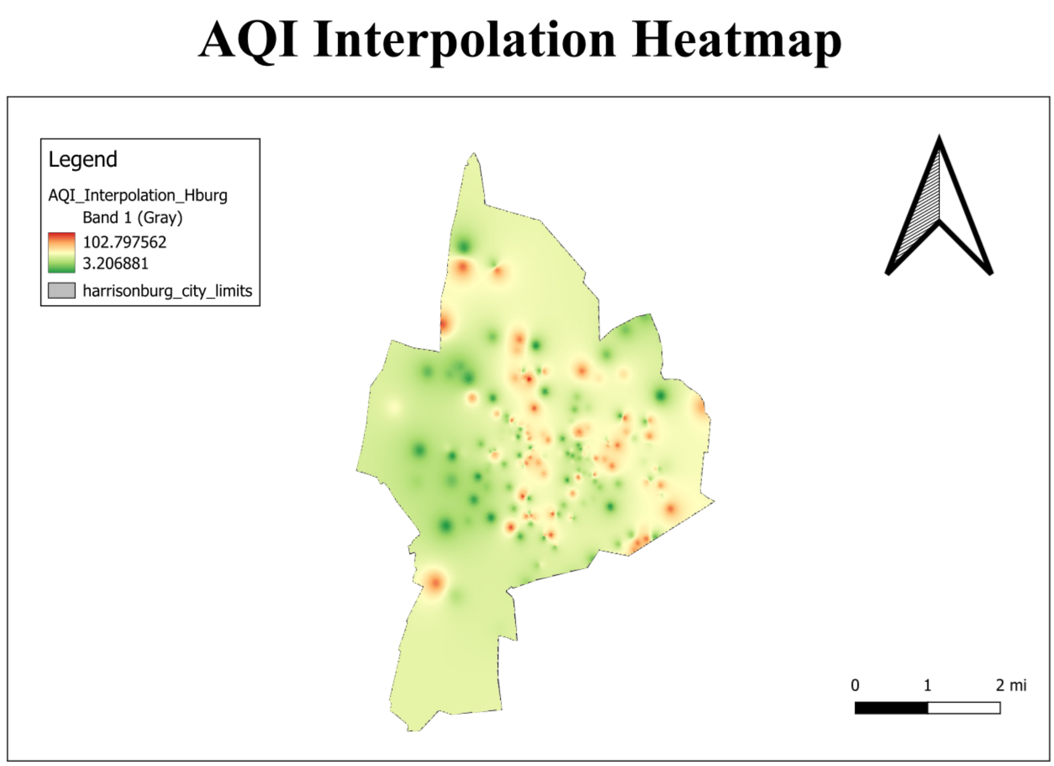

Figure 3: Map 3

3.3 Graphs

Finally, we examine the scatterplots of PM 2.5 correlated to AQI, TempC, and PM 10. It is worth generally noting that, although we would have liked to, we were unable to correlate any variables with ozone (except spuriously) because of the randomly generated data by group five in section three. Additionally, we were unwilling to ignore the outlier data, and so, we settled on discarding the ozone data in exchange for real-world data.

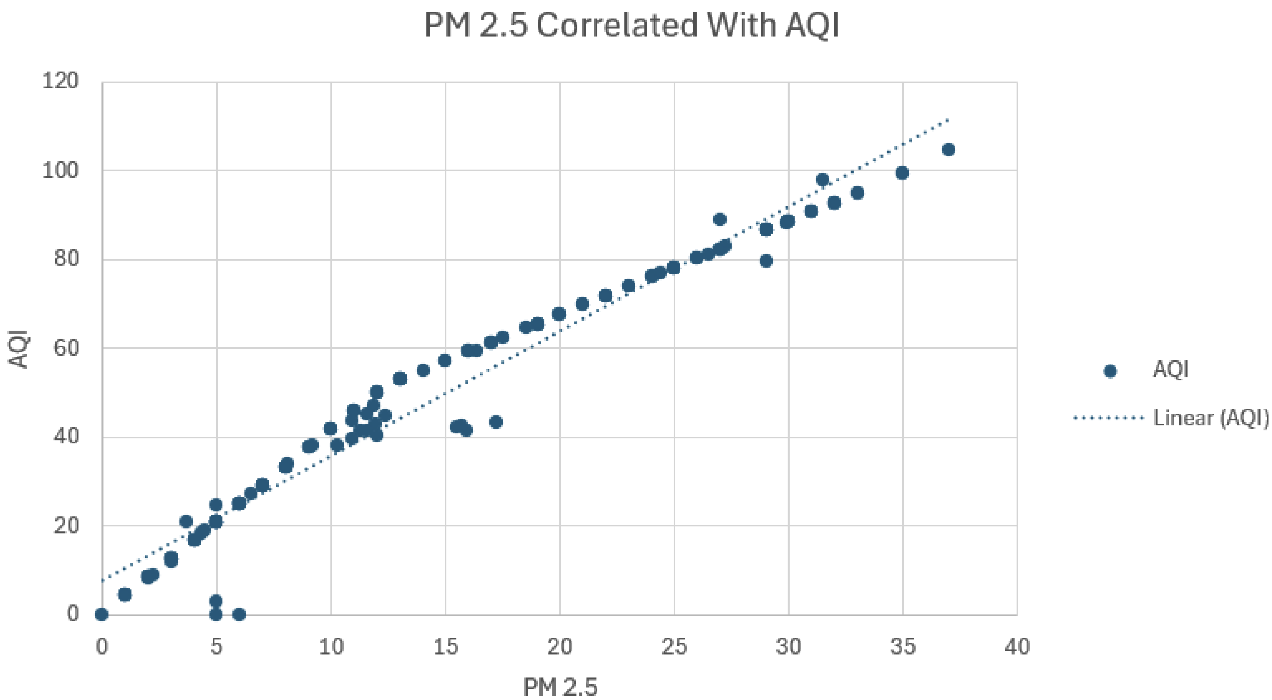

Figure 4: Graph 1

In graph 1, the scatterplot associated to PM 2.5 correlated with AQI, we find that

$$r = 0.965$$

and

$$p \approx 0.$$

This is clearly a nearly linear relationship, although the distribution of residuals will be heteroskedastic (non-normally distributed). If we really wanted to find a good model for PM 2.5 and AQI, we could consider normalizing the distribution of residuals (via implementing a weighted OLS model, a Box-Cox transformation, using robust standard errors, or taking the natural logarithm of AQI data). However, our linear model is close enough to give satisfactory r and p-values which align with our mental framework for what we believe should be true. Accordingly, we do not feel the need to make such adjustments.

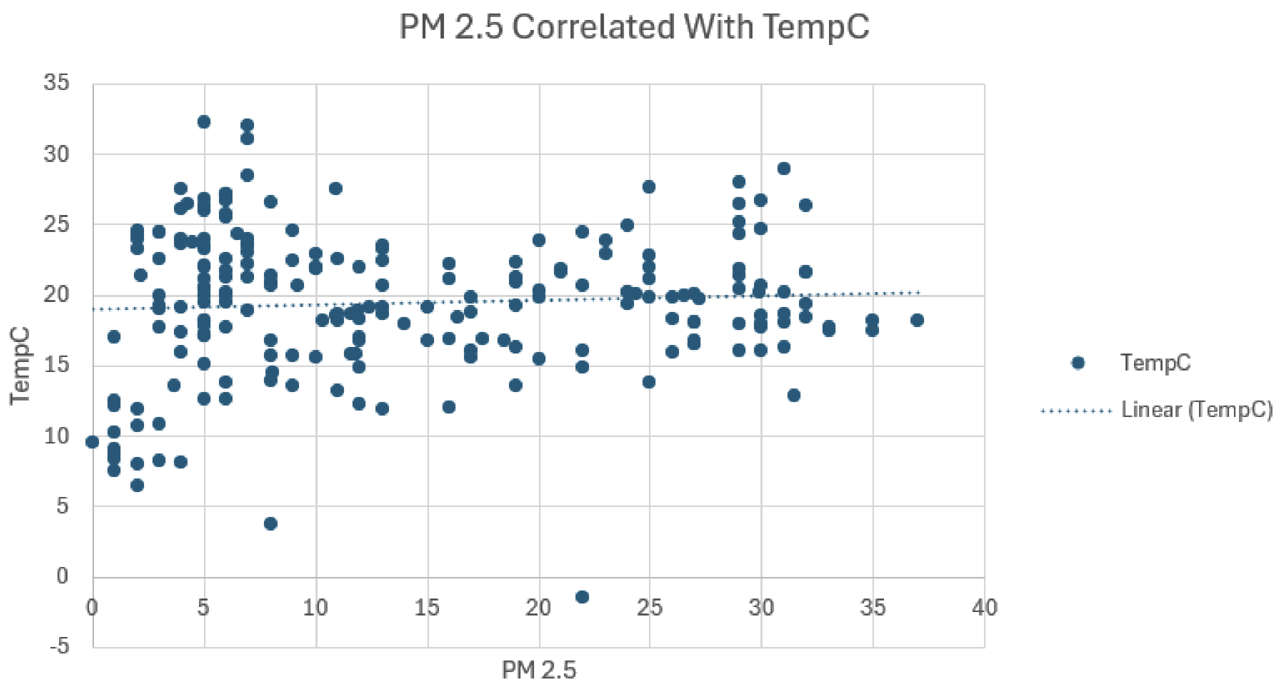

Figure 5: Graph 2

In graph 2, we find horribly varied data across our entire domain of PM 2.5. In this way, we conclude that PM 2.5 levels really have no non-spurious correlation with temperature. This is further justified by our correlation coefficient and p-values:

$$r = 0.051$$

$$p \approx 0.446$$

Alternatively, in graph 3 (on the next page), we find strongly correlated data between PM 2.5 and PM 10, which is exactly what we would expect given the physical model of how PM data is collected. There are some outliers, once again due to group five in section three, however, these randomly generated numbers are not so egregious that they disrupt our linear model fully. In fact, even with these outliers, we obtain,

$$r = 0.973$$

$$p \approx 0$$

as our correlation coefficient and p-value.

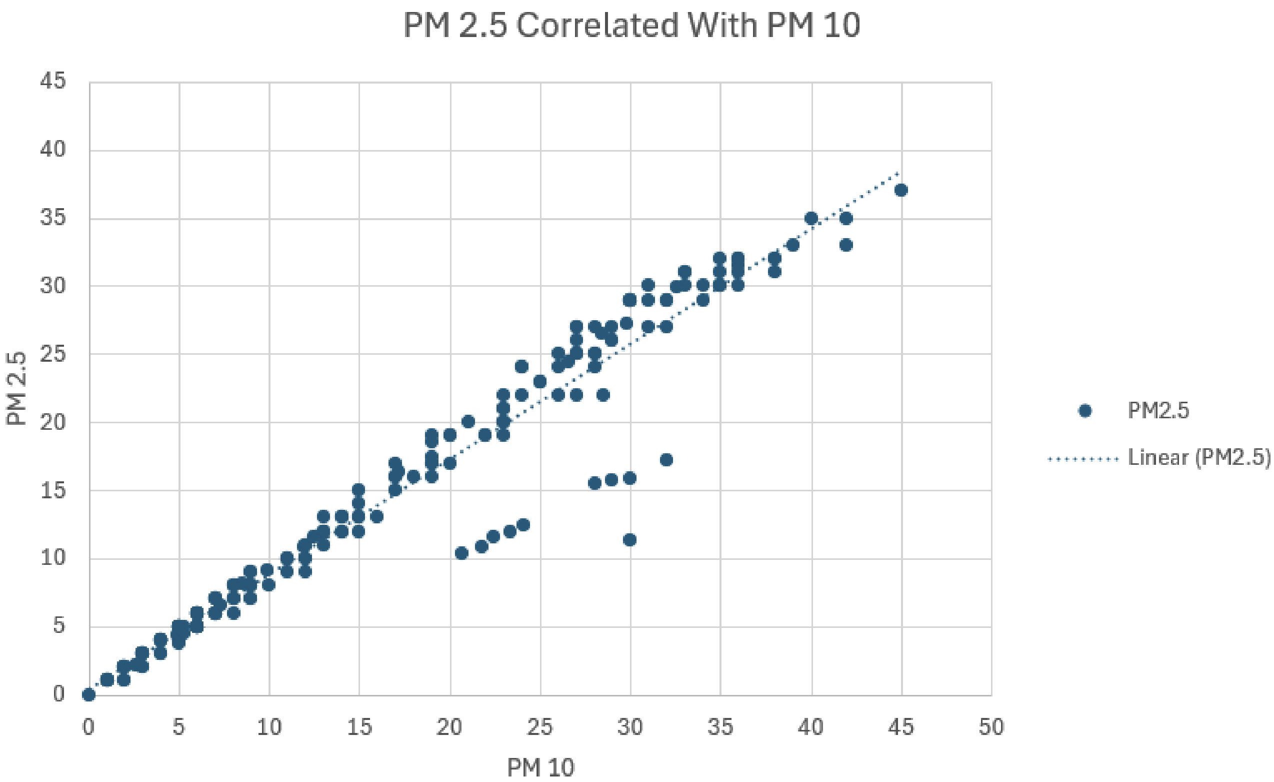

Figure 6: Graph 3

4 Interpretation and Analysis

In this section, we will continue interpreting the data from section three, and we will build an argument to support our conclusions as stated in section 4.6.

4.1 Trends

In section three, we noticed three major tendencies of data collected regarding the air quality in Harrisonburg:

1. The presence of major roads correlates strongly (visually, on maps two and three) with higher PM 2.5 and higher AQI.

2. PM 2.5 is pseudo-logarithmically correlated with AQI; rather, its predictions will have heteroskedastic (non-normally distributed) residuals, and this typically indicates a logarithmic relationship between the variables.

3. Although temperature varies widely on the interval $(-1.5, 32.2)$ degrees Celsius, indicating weather changes, it is not significantly correlated with PM 2.5.

Additionally, although we would have liked to employ the ozone data in our analysis, certain outlier data (seemingly randomly generated) caused us to discard ozone as a potential predictor or dependent variable in our linear regression models.

4.2 Comparison to Air Quality Standards

By the AQI standards given from Du et. Al (2016), given that our median AQI value (from table 2) is 42.2, Harrisonburg's air quality falls in the "good" category.

AQI Category	Index Values	Revised Breakpoints ($\mu g/m^3$, 24-hour average)
Good	0 - 50	0.0 – 12.0
Moderate	51 - 100	12.1 – 35.4
Unhealthy for Sensitive Groups	101 – 150	35.5 – 55.4
Unhealthy	151 – 200	55.5 – 150.4
Very Unhealthy	201 – 300	150.5 – 250.4
Hazardous	301 – 400	250.5 – 350.4
	401 – 500	350.5 – 500

Figure 7: AQI Standards, Du et. Al (2016)

Alternatively, comparing PM 2.5, PM 10, and ozone to the NAAQS table via the EPA's website, since

- (for PM 2.5), our median value of 11 is greater than 9 (the EPA's standard for high PM 2.5), we determine that Harrisonburg has excessive PM 2.5 content.
- (for PM 10), our median value of 13 is less than 150 (the EPA's standard for high PM 10), we determine that Harrisonburg does not have excessive PM 10 content.
- (for ozone), our median value of 0.2 is greater than 0.07 (the EPA's standard for high ozone), we determine that Harrisonburg has excessive ozone content.

From these conclusions, we may suggest policy actions towards reducing PM 2.5 and ozone content (but not PM 10) in the Harrisonburg region, although it is not an entirely imperative matter for ozone content.

4.3 Description of Our Residual Data

When our group collected data, we chose locations based almost solely on convenience; since we both live near JMU campus, our general locus of collection was centered around the city proper, and so there were reasonably many data points collected by other groups in close proximity to our collection sites. This is evidenced by Map 1. That said, as is visible in map one, many of our points where on the southern portion of the locus of collected data, and so our data may have been less geographically predictable than data collected in the center of JMU campus.

Additionally, one of our group members collected data at night on a stormy day, and so some results from that data collection may not be wholly representative of the air quality in general in Harrisonburg. That said, our other group member collected data on a sunny afternoon in good weather, and so our data still proves valuable.

As discussed preceding and following Table 1, the true PM values varied greatly, as did the absolute errors, which had a median of 13.8 micrograms per cubic meter. We interpret both this variation, and larger median of residuals, as being due to one of two potential causes:

1. measurement error, or,

2. differences in measurement due to displaced timing for data collections versus other groups.

We find it unlikely that measurement error was present, due to semi-consistent variation in true PM (and absolute error), and we will discuss the limitation of analyzing time-series data like frequency data in the next section.

4.4 Limitations

Continuing on the notion of time-series data being analyzed with frequency data methods as a problematic form of analysis, we would like to highlight that one of the main flaws in this experiment design comes from non-homogeneous data. Namely, data was collected in an uneven manner both temporally and geographically, as well as both in our group and throughout the collection conducted by the entire class. We can only assume that data can be treated as homogeneous if the ambient environment that the data is collected from is a homogeneous one, without weather conditions or time-of-day dependencies. Namely, the city of Harrisonburg is not such an environment.

In addition to this unfortunate feature of our data, not only can we not view our dataset as frequency data, but we cannot view it as a time-series dataset because our time-partition is not smooth, and we cannot establish suitable lag terms to account for changes in the rate of collection of data. Even if we can mathematically glue such a partition together locally, this local information will certainly not yield a global linear model, or even an ARIMA model, since both the autoregressive terms and the integrated differencing terms will be non-linear (or piecewise linear) in nature. Thus, the two primary tools of conventional data science - econometrics and time-series analysis - are rendered crippled by poor data collection.

Despite these limitations of data collection, however, we can still establish reasonable (multi)-linear models for the real-world processes which we hope undergird our data. Even if a "significant" correlation is spuriously induced by convenient data (for example, see graph 3, correlating PM 2.5 with PM 10), it need not be a spurious correlation in theory. Some may argue though, that an over-reliance on econometric theory may render data useless. It is our opinion that this is - at large - the correct viewpoint. When data stops fitting theoretical models, it signals that either

- the data was collected incorrectly, or,

- something fundamental about the underlying ecosystem has changed, implying that the theoretical framework should be updated.

Up to structural equivalence, it can be quite difficult to distinguish between these two issues, however, in our case, we know that enough data was collected poorly that - if the data supports our theoretical claims, they are then seemingly a vacuous truth, and that if the data fails to support our claims, then we should remain confident in our theory. At any rate, since the data seems to support our conclusions, we will pretend like this truth is anything but vacuous. (This is written somewhat satirically, by a pure mathematician who remains skeptical of data science as a field; in the wise words of Sir Terry Pratchett, "The trouble with having an open mind, of course, is that people will insist on coming along and trying to put things in it." Data science can seemingly be this way quite often...)

4.5 Future Analysis

Based on the limitations discussed in section 4.4, it is clear that a better data collection system is desirable; however, in lieu of this grandiose delusion, we suggest that future work pertain to examining alternative models to the OLS framework, especially for studying the relationship between PM 2.5 and AQI. As stated below graph 1 in section 3.3, one might examine a weighted OLS model with robust standard errors, or apply transformations to the dependent variable (AQI), or apply Box-Cox transformations to the independent variable.

Additionally, our group had hoped to study topological structures within this data, however, we were unable to given certain independent time constraints during the semester.

4.6 Conclusions

In conclusion, we find that the presence of major roads correlates visually with higher PM 2.5 and higher AQI, which are in turn correlated with each other. We loosely examined this correlation and discovered a pseudo-logarithmic relationship between the two variables, which may admit further study. Finally, we determined that although temperature varies widely - indicating weather changes, it is not significantly correlated with PM 2.5 in any meaningful way.

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